

TR-002: Functional Bench Verification Report

Document ID: TR-002

Revision: 1.0

Project: TOMTOM ISR Demonstrator

Verification Method: Test

Verification Level: Subsystem / System

Verification Event: VE-001: Bench Bring-Up

Related Test Procedure: TP-002

1. Purpose

Document the results of executing TP-002 to verify that the TOMTOM Ground Control Station successfully powers on, initializes hardware and software, and performs basic functional bench verification prior to subsystem integration testing.

2. Referenced Documents

- TP-002 Functional Bench Test Procedure
- Verification Workbook
- Applicable Requirements Specification
- Applicable Interface Control Document
- System Requirements Specification

3. Configuration Baseline

Item	Configuration
Computer	ASUS G73S
Operating System	Linux Mint 22.3 Cinnamon (64-bit)
Software Version	QGroundControl V5.0.3-1109-g098330170 (64-bit)
OpenTAKServer	127.0.0.1 https:tak.tomtomtak.com
ATAK Client	Version _____ (If Installed)
Firmware Version	N/A – Standalone Bench Verification
Git Revision	N/A
Procedure Revision	TP-002 Rev. 001
Vehicle Configuration	Not Connected (Bench Verification)
Power Source	External AC Power Supply
USB Devices	Mouse, Flash Drive (as applicable)
Network Connection	Ethernet
Date	20 July 2026

4. Test Personnel

Name	Role
Serena Green	Test Engineer
Alex Green	Observer

5. Test Results

Verification Item	Expected	Actual	Pass/Fail
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Hardware Configuration	Hardware configuration accurately recorded prior to testing.	Hardware configuration was accurately recorded prior to testing procedures.	Pass
Input Voltage	Input voltage is within acceptable operating limits.	AC adapter output measured 19.2 VDC, within manufacturer specifications.	Pass
Ground Control Station Power-On	Ground Control Station powers on successfully.	Ground Control Station computer ASUS G73S powered on successfully with no abnormalities.	Pass
Boot Time	Operating system boots within the required time without errors.	OS booted entirely in 2 minutes/ 120 seconds.	Pass
Operating System Startup	Operating system loads successfully and reaches a usable desktop environment.	OS booted entirely in 2 minutes/ 120 seconds.	Pass
QGroundControl Launch	QGroundControl launches successfully without errors.	QGroundControl launched successfully without any errors.	Pass
OpenTAKServer Launch	OpenTAKServer starts successfully without errors.	OpenTAKServer loaded successfully in Admin/User profiles without errors.	Pass
ATAK Launch (if installed)	ATAK launches successfully and authenticates with OpenTAKServer.	ATAK successfully authenticated with OpenTAKServer using generated PKI certificates.	Pass
CPU Temperature	CPU temperature remains within normal operating limits.	Highest recorded CPU temperature is 36°C	Pass
Cooling Fan Operation	Cooling fan operates normally throughout testing.	Fans observed to continuously produce airflow at exhaust vents. No abnormal noise or vibration was detected during testing.	Pass

Battery Status Display	Battery status is displayed correctly.	Battery status consistently displayed while performing bench testing procedures.	Pass
USB Device Detection	Connected USB devices are successfully detected.	USB devices (mouse, flash drive) were successfully detected at each connection during testing procedures.	Pass
Ethernet Interface Verification	Ethernet interface establishes a stable network connection.	The Ethernet interface was successfully detected and established a stable network connection. Network connectivity was verified through successful access to the OpenTAKServer web interface, server configuration activities, HTTPS service validation, certificate enrollment, and ATAK client communication over the network.	Pass
Storage Device Verification	Storage device is accessible and functioning correctly.	The installed solid-state drive was successfully accessed throughout functional bench verification. The operating system booted normally, software packages were installed, configuration files were created and modified, and applications executed successfully from persistent storage without storage-related errors.	Pass
System Clock Verification	System clock displays the correct date and time.	System clock successfully displays correct date and time.	Pass

Event Logging Verification	System event logging is enabled and operational.	System event logging was enabled throughout functional bench verification. Multiple operating system and service events were successfully recorded and reviewed using journalctl during software startup, service configuration, and troubleshooting activities	Pass
Orderly Shutdown	Operating system shuts down normally without errors.	Operating system successfully shut down normally without any errors.	Pass
Post-Test Hardware Inspection	No abnormal heating, smoke, odor, or hardware damage observed.	No abnormalities or new damage to ASUS G73S were observed during post-test hardware inspection.	Pass

6. Requirement Verification Summary

Requirement	Requirement Statement	Evidence
STK-007	The system shall provide mission data recording.	Fig. 5-C
OPS-005	The operator shall receive mission alerts in real time.	Fig. 5-C
OPS-009	The system shall record mission telemetry.	Fig. 5-C
SYS-FUN-001	The system shall initialize all onboard subsystems prior to flight.	Fig. 3-C
SYS-FUN-005	The system shall provide preflight health monitoring.	Fig. 5-C
SYS-REL-001	The system shall monitor subsystem health.	Fig. 6-C
SYS-REL-002	The system shall log system faults.	Fig. 5-C
SYS-DAT-001	The system shall record mission telemetry during flight.	Fig. 5-C

SYS-DAT-002	The system shall record system faults and warnings during mission operations.	Fig. 5-C
SYS-DAT-003	The system shall assign a unique identifier to each mission data set.	Fig. 5-C
SYS-DAT-004	The system shall preserve recorded mission data following a normal system shutdown.	Not Verified
SYS-DAT-005	The system shall permit authorized users to retrieve recorded mission data after mission completion.	Fig. 5-C
SYS-DAT-006	The system shall store mission timestamps using a consistent time reference.	Fig. 5-C
SYS-REL-005	The system shall perform startup checks of mission-critical subsystems.	Fig. 3-C
SYS-REL-008	The system shall retain fault-log data following an unexpected system restart when technically feasible.	Not verified.
SYS-MNT-005	The system shall permit software updates to the flight-control system without replacement of hardware.	Fig. 6-C

7. Deviations

No procedural deviations were encountered during execution of TP-002 functional test bench. All planned test activities were completed successfully, and the Ground Control Station operated as expected throughout testing.

8. Observations

The Ground Control Station successfully completed functional bench verification without hardware or software malfunctions. Linux Mint, QGroundControl, and OpenTAKServer all launched successfully and remained stable throughout testing. OpenTAKServer successfully authenticated the ATAK client and maintained network connectivity for the duration of the verification. The OpenTAKServer dashboard provided continuous monitoring of system status, hardware information, and resource utilization, while hardware sensor data confirmed that CPU temperature and cooling fan operation remained within normal operating limits. Storage, network, and software subsystems functioned as expected and supported all planned verification activities without interruption.

9. Discrepancies

No discrepancies were identified during TP-002 functional bench verification. All planned test activities were completed successfully, and the Ground Control Station operated as expected throughout testing.

10. Overall Assessment

The Ground Control Station successfully completed functional bench verification and demonstrated stable operation throughout testing. Linux Mint, QGroundControl, OpenTAKServer, and ATAK operated as intended, supporting verification of hardware functionality, software startup, network connectivity, system monitoring, and mission support capabilities. All applicable TP-002 verification activities were completed successfully, no discrepancies were identified, and the Ground Control Station is considered ready to support subsequent subsystem integration and system-level verification activities.

11. Evidence References

Figure	Description
Figure 1-C	OpenTAKServer profile page showing successful ATAK client enrollment and authenticated connection.
Figure 2-C	OpenTAKServer dashboard displaying server status, system information, resource utilization, and operating environment during bench verification.
Figure 3-C	QGroundControl successfully launched and displaying the mission planning interface following application startup.
Figure 4-C	Linux Mint desktop environment successfully loaded following operating system startup.
Figure 5-C	QGroundControl displaying simulated mission operation, telemetry, and system status during functional verification.
Figure 6-C	OpenTAKServer dashboard and Linux hardware sensor output showing normal system operation, CPU temperature, and cooling fan performance during testing.

12. Approval

Role	Name	Signature	Date
Test Engineer	Serena Green	Serena Green	20 July 2026

Reviewer			
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Appendix C: Verification Evidence

The screenshot displays the OpenTAKServer web interface in a Mozilla Firefox browser. The page title is "serenagreen". Below the title, it shows the last login time as "Sat, 18 Jul 2026 15:23:45 GMT" and the email address "serenagreen.eng@gmail.com". A language dropdown menu is set to "English".

The main content area features a table with columns: Callsign, Device, Platform, OS, Phone Number, UID, Version, Last Event Time, and Last Event. The table contains one entry for the user "GUMBALL".

Callsign	Device	Platform	OS	Phone Number	UID	Version	Last Event Time	Last Event
GUMBALL	GOOGLE PIXEL 6A	ATAK CIV	36	7038017325	ANDROID 6caab8229eb05cb3	5.6.0.12 (9c9a5897)[playstore].1769863102 CIV	Sun, 19 Jul 2026 18:56:42 GMT	Connected

The left sidebar contains various navigation options: Dashboard, Map, EUDs, Alerts, CasEvac, Data Packages, Video Streams, Video Recordings, Meshlastic, Missions, Download Truststore, ATAK QR Code, ITAK QR Code, 2FA Setup 2FA, and Dark Mode.

Figure 1. *OpenTAKServer profile page showing successful ATAK client enrollment and authenticated connection.*

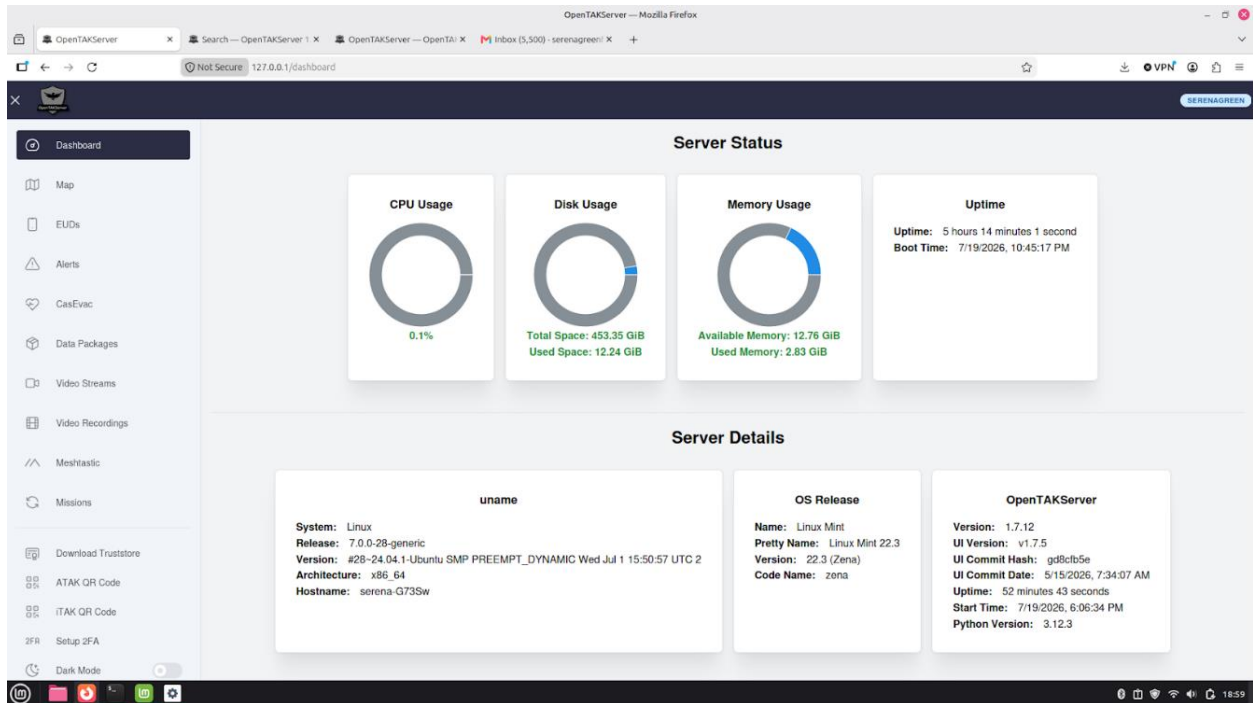


Figure 2. OpenTAKServer dashboard displaying server status, system information, resource utilization, and operating environment during bench verification.

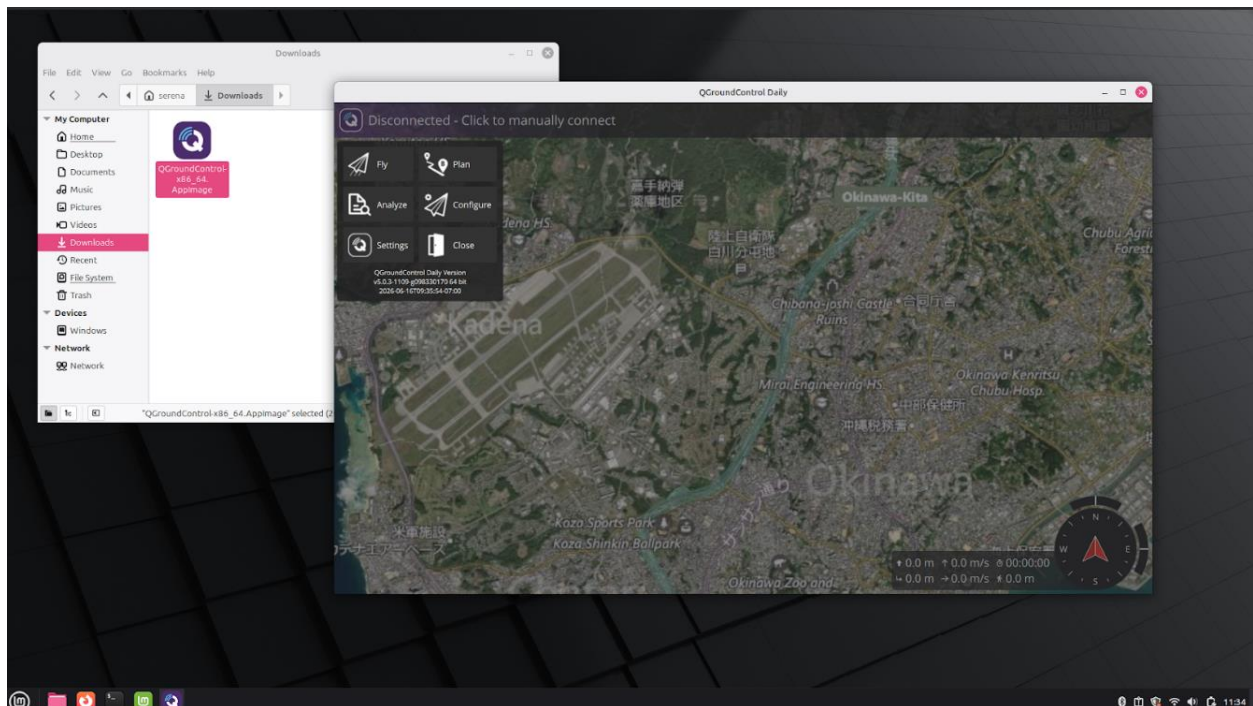


Figure 3. QGroundControl successfully launched and displaying the mission planning interface following application startup.

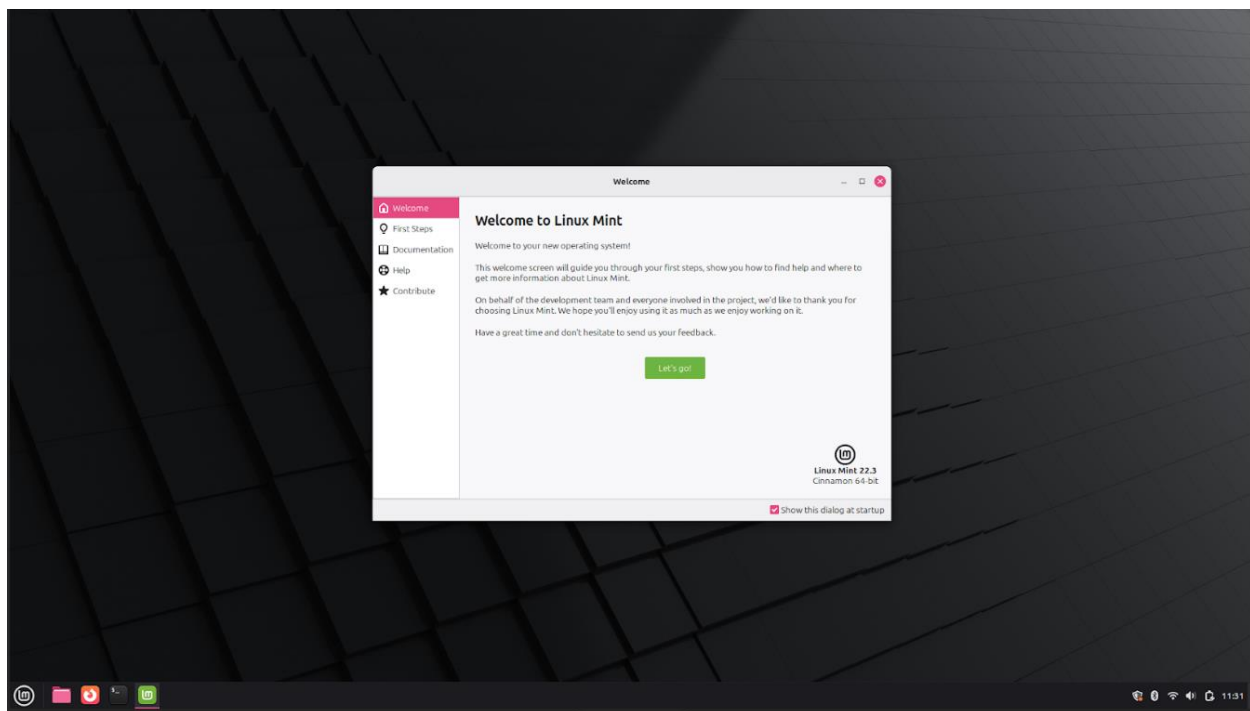


Figure 4. *Linux Mint desktop environment successfully loaded following operating system startup.*

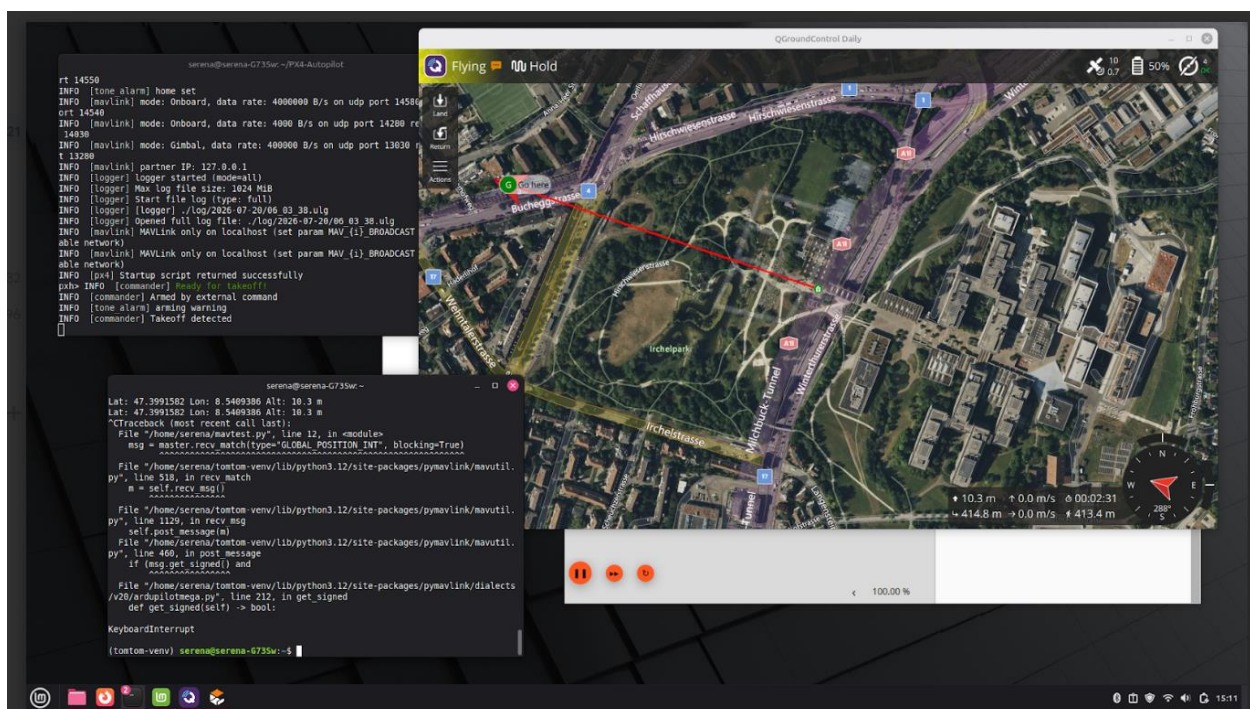


Figure 5. *QGroundControl displaying simulated mission operation, telemetry, and system status during functional verification.*

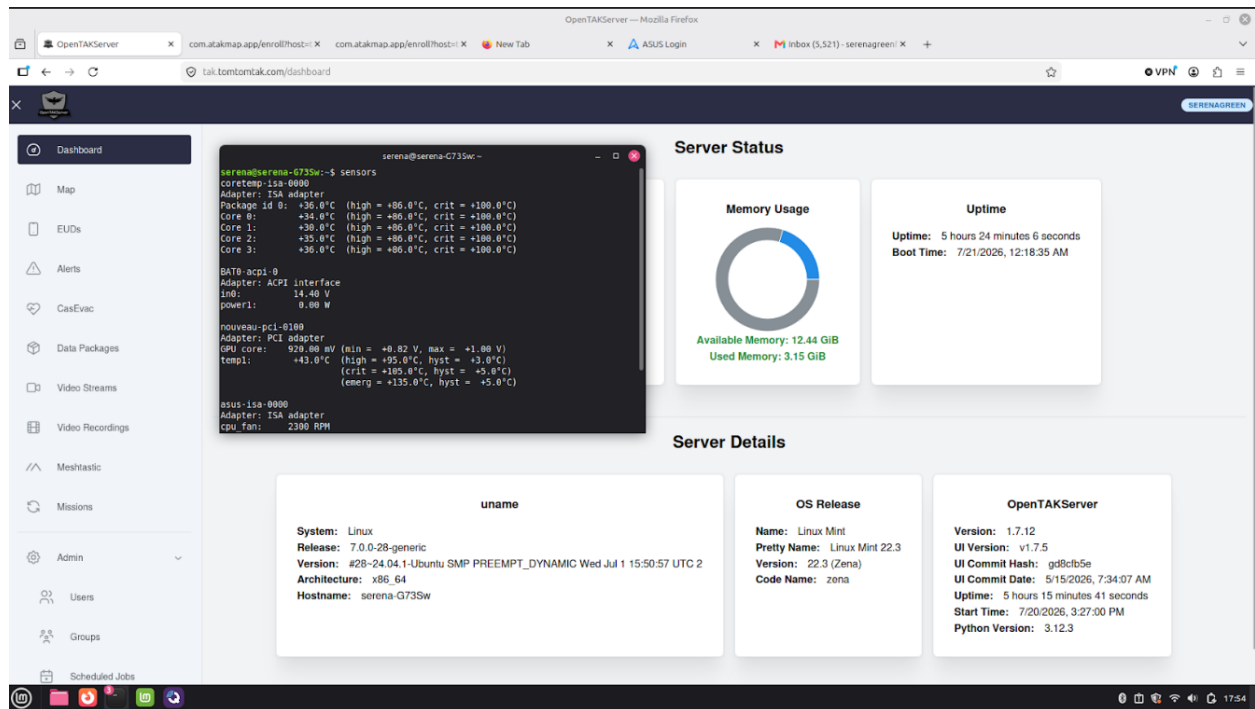


Figure 6. OpenTAKServer dashboard and Linux hardware sensor output showing normal system operation, CPU temperature, and cooling fan performance during testing.